



Do fractions fill the number line?



Class video: https://youtu.be/PprNvG_hwVE

The code leads to the video.

Name: _____ Date: _____

While you watch the video

1. **The bet.** If we mark every fraction on the number line, will any gap be left? Yes / No. Why do you think so?
2. What is $\frac{3}{4}$ as a decimal? And $\frac{1}{3}$? How are their decimals different?
3. Divide 3 by 11 by hand and write down the remainders. What happens?
4. Before the video shows it: what is $0.999\dots$ equal to? Why do you think so?

Practice

1. Write $0.4545\dots$ as a fraction and simplify it.

2. Do $\frac{1}{8}$ and $\frac{1}{12}$ end or repeat? Explain why, looking at the remainders.

3. Write three fractions between $\frac{1}{3}$ and $\frac{1}{2}$.

Keep thinking

If $0.999\dots$ and 1 are the same number, a number cannot be what we write. So what is a number?